Charm Industrial Isometric Standard Project Verification Report

Client	Isometric
Company / GHG Removal Supplier	Charm Industrial
Facility or Project Name	Charm Kansas Bio-Oil Geologic Storage
Facility Address	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses, Port-Cartier, QC G0W 1X0, Canada Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 Bio-oil Injection: Vaulted Deep 7513 KS-14, Hutchinson, KS 67501
Greenhouse Gas Statement ID	Isometric ggs_1JHNY6WK41S0D1QJ
Net Greenhouse Gas Removals	246.73 tonnes
Removal Method	Bio-Oil Geological Storage
Removal Period	June 1 – July 14, 2024 + Oct 29 – Dec 6, 2024
Verification Opinion	Fully meets criteria
Verifier/Validation and Verification Body	Tim Hansen Lily Schacht 350Solutions, Inc.

ISSUED: FEB 21, 2025

CONTENTS

1. Key Project Data 3
1.1. Verification Summary 4
2. Verification Details 5
2.1. Verification Objectives 5
2.2. Verification Team Details 5
2.3. Audit Process Followed 5
3. Technical Aspects 6
3.1. Project Details 6
3.2. Protocols and Modules Applied 7
3.3. Applicability of Applied Protocol and Modules 7
3.4. Project Boundary 7
3.5. Changes from Previous Verifications 7
4. Crediting Details 8
4.1. Project Reporting Period 8
4.2. Crediting Period 9
5. GHG Statement Verification 9
5.1. Accuracy of the Statement 9
5.2. Verification Opinion 9
5.3. Verification Results Summary 10
5.4. Verification Findings 12
6. Revision History 13
Appendix 1: Documents Submitted for Review 15
Appendix 2: Verification Audit Findings and Responses 29
Appendix 3: Verifier Qualifications 31

ISOMETRIC STANDARD VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Lily Schacht Guy Ingram-Hardwick
Removal Method: Bio-Oil Geological Storage		
Report Date: February 21, 2025		
Document No: 350VR-CHARM-ISO2504-122024		
Rev: v1.2		

** primary contact/lead author*

1. KEY PROJECT DATA

TABLE 1. CHARM TECHNOLOGY VERIFICATION SUMMARY

Title	Charm Kansas Bio-Oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses, Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection: Vaulted Deep 7513 KS-14, Hutchinson, KS 67501 37.966035, -97.947845
Reporting Period	June 1 – July 14, 2024 and October 29 – December 6, 2024
Greenhouse Gas Statement ID	Isometric ggs_1JHNY6WK41S0D1QJ
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (mill waste supplier) Geologic Storage Management: Vaulted Deep Regulatory Agency (Injections / Storage): Kansas Department of Health and Environment (KDHE)
Legal Ownership of the Credits	Charm Industrial
Removals in Reporting Period	246.73 tonnes
Removal Method	Bio-Oil Geological Storage in Salt Caverns
Verification Report Date	February 21, 2025
Verification Type & Criteria	Isometric Standard Verification for ongoing deliveries of CDR by Charm Industrial in accordance with the documents used during project validation: • Isometric Standard v1.2 • Bio-Oil Geological Storage Protocol v1.0.1 • Biomass Feedstock Accounting Module v1.1 • Biomass or Bio-oil Storage in Salt Caverns v1.0.2 • Embodied Emissions Accounting v 1.0.2 • Energy Use Accounting v 1.1 • Transportation Emissions Accounting v1.1 (updated since project validation)

1.1. VERIFICATION SUMMARY

350Solutions was contracted to perform a project verification of the greenhouse gas (GHG) statement of Charm Industrial's bio-oil geological storage process for reporting period including June 1 – July 14, 2024, and October 29 – December 6, 2024. Between July 15 and October 28, this project returned to the research and development (R&D) phase while a technical issue was resolved. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast and issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. The validation remains valid for this verification reporting period. Further details and findings can be found in the validation report (https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation).

350Solutions Inc. declares that it is an impartial verification body, approved by Isometric for verifications in the required technical area, free from any conflicts of interest, capable, and qualified to complete this verification for the current reporting period according to the *Isometric Standard V1.2, Bio-oil Geological Storage Protocol v1.0.1, Biomass Feedstock Accounting Module V1.1*, *Bio-oil Storage in Salt Caverns Module v1.0, Embodied Emissions Accounting v1.0.2, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.1.1*.

350Solutions affirms that Charm has the appropriate equipment, procedures, and protocols in place and documented to quantify GHG removal through the injection of bio-oil into salt caverns via permitted Class V wells in accordance with the requirements of the applicable Isometric standards, protocols and modules.

The carbon removals for the reporting period are provided in Table 2. These values have been verified by 350Solutions.

TABLE 2. VERIFIED TOTAL REMOVALS BY CHARM INDUSTRIAL FOR REPORTING PERIOD ENDING DEC 6, 2024

Removal ID	Supplier reference ID	Date removal started	Date removal completed	Discounted CO_2e net removed, rounded (kg)
rmv_1JH8C24TJ1S0DF62	745	2024-11-06	2024-11-06	470
rmv_1JH8C6JYY1S0V2F5	746	2024-11-15	2024-11-15	3770
rmv_1JH8CA5PC1S0V4ST	747	2024-11-19	2024-11-19	16680
rmv_1JH8CGQE71S08P4D	748	2024-11-20	2024-11-20	13530
rmv_1JH8CSVNQ1S04KFS	749	2024-11-21	2024-11-21	29350
rmv_1JH8CVAV01S0AW2A	750	2024-11-26	2024-11-26	46330
rmv_1JH8CYAMD1S09XV9	751	2024-12-04	2024-12-04	44870
rmv_1JH8D0DQ91S0FRK3	752	2024-12-05	2024-12-05	46500
rmv_1JH8D26801S0QPCS	753	2024-12-06	2024-12-06	45230
TOTAL REMOVALS				**246730 kg (246.73 tonnes)**

The quantity of 246.73 tonnes of net carbon removals has been verified with a reasonable level of assurance and are eligible for crediting under the referenced Isometric Standard and Protocol.

Verifier Information		
Verification Body	**Verifier**	**Verification ID No.**
350Solutions, Inc.	Lily Schacht	VS-Charm-ISO2504

Lily Schacht Digitally signed by Lily Schacht Date: 2025.02.21 09:51:58 -05'00'

Signed: Lily Schacht (Verifier)

Guy Hardwick Digitally signed by Guy Hardwick Date: 2025.02.21 15:06:37 Z

Guy Ingram-Hardwick (QA)

2. VERIFICATION DETAILS

2.1. VERIFICATION OBJECTIVES

A series of verification activities was conducted by 350Solutions to independently verify the claimed removals of the GHG statement submitted by Charm Industrial for operations conducted at the Kansas Bio-oil Geologic Storage facility for the time-period June 1 – July 14, 2024, and October 29 – December 6, 2024. Between July 15 and October 28, this project returned to the research and development (R&D) phase while a technical issue was resolved. The verification was conducted following the specifications of *Isometric Standard V1.2, Bio-oil Geological Storage Protocol v1.0.1, Biomass Feedstock Accounting Module V1.1*, *Bio-oil Storage in Salt Caverns Module v1.0, Embodied Emissions Accounting v1.0.2, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.1.1*. The verification is based on the provision and fair presentation by Charm Industrial and Isometric of a complete and accurate GHG Statement in accordance with the criteria (Table 1, Section 3.2) which is confirmed and verified by 350Solutions based on supporting evidence. The verification was conducted with a reasonable level of assurance and discrepancies between the claimed removals of the project and the removals estimated by the verification process for the reporting period are less than the materiality threshold of 5%.

2.2. VERIFICATION TEAM DETAILS

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020 for completion of ISO 14034 Technology Verifications. 350Solutions Technical Lead for the Charm verification is Tim Hansen. Mr. Hansen is a registered Professional Engineer with over 20 years of experience in the evaluation and verification of novel cleantech and carbon technologies. Mr. Hansen has led verifications of carbon removal technologies for multiple registries, including Isometric, and has experience in the development and testing of biomass conversion technologies, including gasification and pyrolysis. The verification report was written by Lily Schacht, a Senior Carbon Removal Verification Engineer, with 4 years of experience developing and verifying low-carbon and carbon removal technologies. Complete verifier qualifications are attached in Appendix 3.

2.3. AUDIT PROCESS FOLLOWED

The specific verification activities conducted are summarized in Table 3. The list of all files that were reviewed is provided in Appendix 1. The list of findings from the verification can be found in Appendix 2. Audit results are documented using 350's Compliance Checklist, which includes requirements from the Isometric standard, Bio-oil geologic Storage protocol, and referenced modules.

TABLE 3. VERIFICATION ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
Feb 10, 2025	Review of reporting period modifications to PDD	- Review of updated PDD and GHG Statement	V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx 10_29-11_6 GHG Statement Report v1.6 (Charm v1.1).docx.pdf
Feb 10-19, 2025	Review of current Validation and GHG Statement Applicability	- Review of Validation Report - Review of prior Verification Report - Identification of major and minor process steps, measurements, and calculations - Review of Standards/Protocols/Modules required for Verification - Generation of a checklist of verification issues, recommendations, and opportunities for improvement	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1 Charm Verification Report 2024-07-26 Charm_ISO2405_VRFINALV1.2 V 1.7 Charm Industrial G.P. Bio Oil Sequestration PDD.docx.pdf PDD Change-Log.xlsx 10_29-11_6 GHG Statement Report v1.6 (Charm v1.1).docx.pdf Verifier Guidance_10_29-12_6 Reporting Period.pdf Isometric Standard V1.2, Bio-oil Geological Storage Protocol v1.0.1, Biomass Feedstock Accounting Module V1.1, Bio-oil Storage in Salt Caverns Module v1.0, Embodied Emissions Accounting v1.0.2, Transportation Emissions Accounting v1.1, and Energy Use Accounting v1.1.1.
Feb 14-21, 2025	Verification of Removals	- Verify changes to the system - Review of measurements and calculations - Review of durability/permit requirement monitoring data - Review of GHG statement - Identification of initial findings and follow-up request for information - Write verification report	File List provided in Appendix 1. Initial Findings in Appendix 2.

3. TECHNICAL ASPECTS

3.1. PROJECT DETAILS

Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/ Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill. Bio-oil is shipped by tanker truck or rail car to a Charm facility in El Dorado Kansas. At this facility, Charm pretreats the bio-oil through the addition and sparging of waste brine (salt) and liquid caustic soda (LCS) to adjust the pH of the bio-oil and the salt content or conductivity to meet the requirements of the injection facility permit for storage in salt caverns as specified by the Kansas Department of Health and Environment (KDHE) and injection partner Vaulted Deep's Class V injection permit. Following treatment, samples are taken for CHN analysis (carbon content), total acid number (TAN), and density testing and the oil is transported to the Vaulted Deep injection facility.

Once successfully pre-treated to ensure EPA Underground Injection Control (UIC)/KDHE compliance, including measurement of pH and conductivity both within allowable permitted ranges, the oil is trucked to Vaulted Deep's Great Plains Facility in Hutchinson, KS for injection into an underground salt

cavern. At that point, the oil is offloaded and Vaulted takes custody of the product and responsibility for emplacing it into the storage cavern. The waste is not stored on site, instead it is processed and injected on the same day as it arrives on site. Therefore, emissions associated with waste storage at the facility are avoided. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment are detailed in their own PDD, which has been previously validated [1].

3.2. PROTOCOLS AND MODULES APPLIED

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard v1.2 [2]
- Bio-Oil Geological Storage Protocol v1.0.1 [3]
- Biomass Feedstock Accounting Module v1.1 [4]
- Biomass or Bio-oil Storage in Salt Caverns v1.0.2 [5]
- Embodied Emissions Accounting v 1.0.2 [6]
- Energy Use Accounting v 1.1 [7]
- Transportation Emissions Accounting v 1.1 [8]

Transportation Emissions Accounting Module v1.0 was used in the initial validation and previous verifications of this project. The module was updated (V1.1) to allow for the use of Book and Claim Units (BCUs) to substitute for some, or all, or project fuel usage. No other changes were made to the module.

3.3. APPLICABILITY OF APPLIED PROTOCOL AND MODULES

The Charm project takes mixed woody waste, converts it to bio-oil, which they pretreat and inject into salt caverns. The mixed woody waste was assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and was found to be eligible. As the biomass feedstocks have been deemed eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in underground salt caverns makes both the Bio-oil Geological Storage Protocol and Biomass/Bio-oil Storage in Salt Caverns Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.4. PROJECT BOUNDARY

The boundaries of the project include the supply wood waste to the biomass processing site (AECN), the conversion of the biomass to bio-oil via AECN's pyrolysis process, transport of the bio-oil to Charm for preprocessing at its El Dorado, Kansas facility, completion of pretreatment, shipment to the Vaulted Deep Great Plains facility, and injection of the bio-oil in the salt caverns. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary.

3.5. CHANGES FROM PREVIOUS VERIFICATIONS

Compared to prior verifications, Charm has made several modifications to the process which have been identified in the GHG statement and supporting documentation. The following modifications in project activities were made, in addition to updates to delivery data and monitoring data for the reporting period:

- Increased separation of bio-oil into light and heavy fractions prevented reliable blending during pre-treatment. As a result, the average carbon (C) content of the bio-oil could not be reliably determined using the sparged oil, as had been done previously. The process was modified during the R&D period (July 15 – October 29) and operations resumed. Removals have not been claimed for the June 1 – July 14 injections because of uncertainty in the C content. Operating emissions from June 1 – July 14 have been included in the net CDR calculation for this reporting period. Additionally, all amortized equipment embodied emissions from June 1 – December 6 are accounted for, even though removals were not claimed for the period from July 1 – October 29. This is appropriately conservative because the equipment continued to be used during the R&D period. This plan was approved by Isometric on July 29, 2024.
- Disposal of bio-oil was required at two points during this reporting period:
    - A contained bio-oil spill occurred at the beginning of June, which required the removal of bio-oil contaminated turf and disposal in a landfill.
    - During the June 1 – July 15 period, overly viscous bio-oil was disposed of in a landfill. Emissions associated with transport and disposal have been included in the net CDR calculation.

  The emissions associated with transport and disposal have been included in the net CDR calculation. The emissions factor for landfilling bio-oil was calculated as the average of the emissions factors for landfilling of wood waste (expected lower bound) and landfilling of food waste (expected upper bound). Additional details on the rationale for this approach can be found in the GHG Statement.
- BCUs were used to offset the emissions associated with fossil diesel usage for transportation with emissions associated with biodiesel usage. The transportation emissions accounting is compliant with the requirements in Section 4 of Transportation Emissions Accounting Module v1.1
- The El Dorado, Kansas pre-processing site has been connected to grid power. Updates to the net CDR calculation have been made accordingly.
- Vaulted used a new cavern, AR-9, for injections of Charm's bio-oil during this reporting period. The cavern is part of the existing permit and cavern system documented in both Charm's and Vaulted's PDDs.
- Charm purchased 66 275-gallon tote tanks, 63 330-gallon tote tanks, 3 storage tanks and tank bases, 8 sliding vane pumps, a diesel storage tank, and a pressure washer for the pretreatment facility in El Dorado, KS. All materials used have been accounted for in the embodied emissions for the equipment, which are allocated to removals appropriately and in accordance with the Isometric protocol requirements.
- Testing approach for C content has changed from three samples per production batch to one sample per injection. This is compliant with the Bio-oil Geologic Storage Protocol and was confirmed by Isometric.

4. CREDITING DETAILS

4.1. PROJECT REPORTING PERIOD

This verification considers injections of bio-oil by Charm during the period June 1 – December 6, 2024 with reliable C content data. From June 1 – July 14, injections occurred but did not have reliable measurements of C content. From July 15 – October 29, the project was in an R&D phase and no creditable removals took place. The reporting period is based on the actual injection dates, with bio-oil

production, transportation, monitoring, and related activities potentially occurring outside of this reporting period.

4.2. CREDITING PERIOD

The 5-year crediting period for the Charm Kansas Bio-oil Geologic Sequestration Project began March 1, 2024 and continues for five years, ending February 28, 2029. All claimed removals in this reporting period fall within the current crediting period.

5. GHG STATEMENT VERIFICATION

5.1. ACCURACY OF THE STATEMENT

The values represented in the GHG statement have been evaluated, cross-referenced with the presented evidence, and confirmed. Confirmation of the GHG statement has been based on the following efforts:

- Duplication of calculations of total bio-oil and CO_2 equivalent injected for Removals 749 and 752 (supplier reference IDs), including:
    - Calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content, addition of consumables (salt and LCS)
    - Review and calculation of average carbon content from raw data (lab reports)
    - Confirmation of transportation distances for tanker and rail car deliveries from AECN to El Dorado, KS
    - Confirmation of weigh scale raw data for deliveries
    - Re-calculation of transportation associated emissions for such transport.
    - Confirmation of raw data for fuel usage at Charm pre-treatment facility
    - Confirmation of electricity usage at Charm pretreatment facility and Vaulted Deep injection facility from raw data
    - Confirmation of electricity emissions factors for electricity for the Kansas region and recalculation of the emissions and confirmation
- Review of monitoring data as required for salt cavern injection / emplacement and post injection monitoring to ensure ongoing compliance with Class V permit

As with the previous verification, some minor discrepancies were noted due to rounding errors both internally at Charm and in communication between Charm's and Isometric's data platforms. Charm and Isometric are working on fixing the issues. The estimated impact of the total discrepancies is 0.02%, so this is not a material issue.

5.2. VERIFICATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the greenhouse gas removal statement indicates that the claims fall within the 5% materiality threshold for greenhouse gas emissions and removals as required by the Isometric Standard:

☒ ***Meets the requirements of the Isometric Standard V1.2.0, Biomass Geological Storage Protocol V1.0.1, Biomass Feedstock Accounting Module V1.1, and Biomass or Bio-oil Storage in Salt Caverns Module v1.0.2 for Verification***

☐ ***Meets the requirements with minor revisions of the Isometric Standard V1.2.0, Biomass Geological Storage Protocol V1.0.1, Biomass Feedstock Accounting Module V1.1, and Biomass or Bio-oil Storage in Salt Caverns Module v1.0.2 for Verification***

☐ ***Does Not Meet the requirements of the Isometric Standard V1.2.0, Biomass Geological Storage Protocol V1.0.1, Biomass Feedstock Accounting Module V1.1, and Biomass or Bio-oil Storage in Salt Caverns Module v1.0.2 for Verification***

5.3. VERIFICATION RESULTS SUMMARY

A summary of verification results and compliance with relevant requirements of the *Isometric Standard, Biomass Geological Storage Protocol, Biomass Feedstock Accounting Module*, and *Biomass or Bio-oil Storage in Salt Caverns Module* are summarized below.

TABLE 4. VERIFICATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	N	Acceptable – No change since validation
2.5.1	*Boundaries*	N	Acceptable – No change since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality - Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality - Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation
2.5.5	*Default Emission Factors, Proxies, and Models*	N	Acceptable – No change since validation. See findings (5.4) for forward action requests and recommended updates
2.5.6	*Common Calculation Factors*	N	Acceptable – No change since validation
2.5.7	*Uncertainty*	N	Acceptable – No change since validation
2.5.8	*Durability and Monitoring*	N	Acceptable – No change since validation
2.5.9	*Risk of Reversal*	N	Acceptable – No change since validation
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	N	Acceptable – No change since validation
2.5.11	*Life Cycle Assessment (LCA) Policies*	N	Acceptable – No change since validation. LCA calculations updated for current reporting period.
3.1	*Ownership*	N	Acceptable – No change since validation
3.2	*Documentation*	N	Acceptable – No change since validation
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV platform.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete and all removals have been verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No change since validation
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	Acceptable – Maintenance of well operation permits meets requirements and ongoing permit reporting requirements completed.
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	N	Acceptable – No change since validation
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since validation
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since validation
3.0	*Counterfactual Definition*	N	Acceptable – No change since validation
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since validation
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since validation
Biomass or Bio-oil Storage in Salt Caverns Module			
1.0, 2.2	*Salt Cavern Characterization – Cavern Dimensions*	Y	Acceptable – Cavern not used until Charm's injections in November. Vaulted Report for SW-12 dated Jan 24, 2024 applies to AR-9.
1.0, 2.2	*Salt Cavern Characterization – Integrity*	Y	Acceptable – See above
1.0, 2.2	*Salt Cavern Characterization – Permeability*	Y	Acceptable – See above
1.0, 2.2	*Salt Cavern Characterization – Porosity & permeability*	Y	Acceptable – See above, also documented in "Faults and Interbeds_Fill (AR-9)" provided for this verification.
1.0, 2.2	*Salt Cavern Characterization – Subsurface Features*	Y	Acceptable – See above
1.0	*Salt Cavern Characterization – Temperature, pH, conductivity/Cl- conc.*	Y (Required Monitoring Updated)	Acceptable - All required parameters measured and are within the required limits.
1.0	*Salt Cavern Characterization – Brine TOC*	Y (Required Monitoring Updated)	Acceptable – Brine TOC measured prior to emplacement, as required by Class V permit. Confirmed via updated TOC data.
1.0	*Salt Cavern Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – No change since validation
1.0, 3.1.3	*Salt Cavern Characterization – Surface Subsistence*	Y	Acceptable – Vaulted Report for SW-12 dated Jan 24, 2024 applies to AR-9.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
1.0	*Salt Cavern Characterization – Displaced Brine Injection location*	Y	Acceptable – Location changed to cavern AR-9
1.0, 2.3	*Emplacement Infrastructure*	N	Acceptable – No change since validation
1.0	*Emplacement Monitoring*	Y (Required Monitoring Updated)	Acceptable – monitoring completed and updated as required by Class V well permit
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation
2.1	*Permit for Operation*	N	Acceptable – No change since validation
3.1.1	*Injection Pressure*	Y (Required Monitoring Updated)	Acceptable – Pressure is continuously monitored and reported to KDHE monthly. Updates provided as required.
3.1.1	*Volume monitoring*	Y (Required Monitoring Updated)	Acceptable – Monitoring of injected material measured daily and reported monthly, confirmed by sonar quarterly. Updates provided as required.
3.1.1	*Injectate monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.1	*Sampling Frequency*	Y	Acceptable – No change since validation. Three samples collected for every injection batch. All sampling data provided and updated as required
3.1.1	*Safety Monitoring*	N	Acceptable – No change since validation
3.1.2	*Mechanical Integrity*	N	Acceptable – No change since validation
3.1.3	*Well Pressure Monitoring*	Y (Required Monitoring Updated)	Acceptable – Well pressure measured continuously with safety interlock. Daily pressure reported to KDHE monthly. Updated monitoring data provided as required.
3.1.3	*Sonar Survey*	N	Acceptable – No change since validation
3.1.3	*Fill Volume*	Y (Required Monitoring Updated)	Acceptable – Cavern volume tracked based upon volume of emplaced material. Estimate volume reported monthly, verified by sonar quarterly. Updated monitoring data provided as required.
3.1.3	*Return Brine Monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change since validation.
3.3	*Reversal Risk*	N	Acceptable – No change since validation
3.3.1	*Attribution of Reversals*	N	Acceptable – No change since validation

5.4. VERIFICATION FINDINGS

Assessment of the greenhouse gas removal statement identified several initial findings (Appendix 2). All findings, primarily associated with missing supporting evidence, have been addressed and closed.

Forward Action requests (FAR) and recommendations are also provided in Appendix 2 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of

potential issues to be monitored by verifiers in future verifications. Forward Action requests and opportunities for improvement are summarized below for future verification reference:

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS

ID No	Type	Finding / Issue	Source	Protocol Reference	Action
1	Forward Action Request	Charm injection site (Vaulted Deep operated) is installing new gas monitoring equipment to better identify gas production at the wellhead / brine holding tanks. Updated information should be provided.	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	Provide specifications, design, installation information, calibrations, and operational data for the new gas monitoring equipment when it becomes available and for relevant reporting periods
2	Forward Action Request	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing.	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1		Please provide AECN testing report. Note, not applicable for this reporting period, but should be updated when possible
4	Forward Action Request	Review and update the KS grid electricity EF annually based on changes to grid mix and any eGrid updates		7.4.3	Please review and update as needed going forward
3	Forward Action Request	Biomass feedstock sourcing was confirmed at validation. Because of its critical nature for compliance, we feel it should be confirmed for each verification reporting period - demonstrating no change in feedstock sources since validation		7.2	Please provide confirmation of biomass feedstock sourcing and that it is not different from validated sources (mill wastes) for verifications in the future
8	Recommendation	Transportation emissions utilize "the average “deadhead” or unloaded journey distance for a vehicle of this type according to USDA analysis". This is sufficient for accounting for return trips, but it may be preferable or more appropriate to use return trip distance if the trip route is known (i.e. a full return trip if applicable).		Bio-oil Geologic Storage 7.4.3 (Eq7)	Consider requesting return trip information from biomass delivery and bio-oil delivery transportation services and account for return trip accurately if appropriate

6. REVISION HISTORY

Original date of issue: February 20, 2025

Version	Date Issued	Noted Changes
V1.0	February 20, 2025	N/A Initial Version
V1.1	February 20, 2025	Reporting period dates fixed

V1.2	February 21, 2025	Removal dates fixed

References

[1] "Vaulted Deep Great Plains Facility Organic Waste Sequestration: Isometric PDD." Jan. 23, 2025. Accessed: Feb. 19, 2025. [Online]. Available: https://storage.googleapis.com/isometric--production--eu--public-documents/spl_1H43SYY631S0Z8YR/prj_1HHYZFVGW1S044ZY/pjd_1JJSMS9FS1S0MF1M/pdd.pdf

[2] Isometric, "Isometric Standard v1.2.0." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/standard/1.2.0

[3] Isometric, "Bio-oil Geological Storage V1.1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/bio-oil-geological-storage/1.1

[4] Isometric, "Biomass Feedstock Accounting V1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/biomass-feedstock-accounting/1.1

[5] Isometric, "Biomass or Bio-oil Storage in Salt Caverns V1.0.4." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/salt-cavern-storage/1.0

[6] Isometric, "Embodied Emissions Accounting v1.0.3." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/embodied-emissions/1.0

[7] Isometric, "Energy Use Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/energy-use-accounting/1.1

[8] Isometric, "Transportation Emissions Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/transportation

See Appendix 1 and Table 3 for list of specific files reviewed during the verification.

APPENDIX 1: DOCUMENTS SUBMITTED FOR REVIEW

ggs_1JHNY6WK41S0D1QJ-sources (1).zip
gss_1JHNYFQ0Z1S00PK7-OLD.xlsx
gss_1JKRTCCBV1S0G894_Current.xlsx
gss_1JKRTCCBV1S0G894_Current_350Ver_2-17-2025.xlsx
10_29-11_6 GHG Statement Report v1.6 (Charm v1.1).docx.pdf
Measurement Discrepancy Guidance.pdf
Verifier Guidance_ 10_29-12_6 Reporting Period.pdf
2024 AR-09 Final Report 8-27-24.pdf
ACS March 2024 Facility Survey.pdf
ACS Permit KS-05-155-002 SIGNED.pdf
AR-09 Brine Hydrostatic Test Report 8-27-24.pdf
AR9 Cavern Docs Summary.xlsx
Charm requirements.docx
Faults and Interbeds_ Fill (AR-9).docx
200053 Vaulted Integrated Evidence.pdf
AR-09 Initial Brine TOC 11-6-2024.pdf
June_July Hazwaste Emissions Summary_EF test.xlsx
Bio-Oil Open Dump EF.docx
Wood and FoodWaste Ref EFs.docx
BO Contam Gravel EF (1).docx
Waste Gravel EF.docx
200053 MRV Ops Notes.xlsx
200053 Vaulted Integrated Evidence.pdf
SR24-07536 Batch 200053 MRV(CHN + H2O).pdf
SR24-07538_0000197921.PDF
200054 MRV Ops Notes.xlsx
200054 Vaulted Integrated Evidence.pdf
BATCHES 200054-200057 MRV (TAN + Density) .pdf
BATCHES 200054-200057 MRV(CHN+H2O).pdf
200055 MRV Ops Notes.xlsx
200055 Vaulted Integrated Evidence (Truck 1).pdf
200055 Vaulted Integrated Evidence (Truck 2).pdf
BATCHES 200054-200057 MRV (TAN + Density) .pdf
BATCHES 200054-200057 MRV(CHN+H2O).pdf
200056 MRV Ops Notes.xlsx
200056 Vaulted Integrated Evidence.pdf
BATCHES 200054-200057 MRV(CHN+H2O).pdf
BATCHES 200056-200057 MRV(TAN + Density) .pdf
185-9511.pdf
185-9512.pdf
200057 MRV Ops Notes.xlsx
BATCHES 200054-200057 MRV (TAN + Density) .pdf

BATCHES 200054-200057 MRV(CHN+H2O).pdf
BATCHES 200056-200057 MRV(TAN + Density) .pdf
185-9541.pdf
185-9543.pdf
200058 MRV Ops Notes.xlsx
BATCH 200058-200061 MRV(TAN + Density).pdf
SR24-08436_0000199763.PDF
185-9616.pdf
185-9617.pdf
200059 MRV Ops Notes.xlsx
BATCH 200058-200061 MRV(TAN + Density).pdf
SR24-08436_0000199763.PDF
185-9633.pdf
185-9634.pdf
200060 MRV Ops Notes.xlsx
BATCH 200058-200061 MRV(TAN + Density).pdf
SR24-08436_0000199763.PDF
185-9642.pdf
185-9643.pdf
200061 MRV Ops Notes.xlsx
BATCH 200058-200061 MRV(TAN + Density).pdf
SR24-08436_0000199763.PDF
045T2R2.pdf
046T2R2.pdf
047T2R2.pdf
069T1R0.pdf
070T1R0.pdf
071T1R0.pdf
072T1R0.pdf
073T1R0.pdf
074T1R0.pdf
075T1R0.pdf
076T1R0.pdf
077T1R0.pdf
078T1R0.pdf
079T1R0.pdf
080T1R0.pdf
081T1R0.pdf
BOL Summary.xlsx
June_July Hazwaste Emissions Summary.xlsx
June_July Scrapped BO Disposal.pdf
06062024 Spill Disposal p1.jpg
06062024 Spill Disposal p2.jpg

06072024 Spill Disposal p1.jpg
06072024 Spill Disposal p2.jpg
06112024 Spill Disposal.jpg
06122024 Spill Disposal.jpg
06132024 Spill Disposal.jpg
06142024 Spill Disposal.jpg
06172024 Spill Disposal.jpg
06202024 Spill Disposal.jpg
06212024 Spill Disposal.jpg
Screenshot 2024-12-06 at 2.13.30 PM.png
Screenshot 2024-12-06 at 2.48.56 PM.png
Bio-Oil Contaminated Gravel Disposal EF Calc Sheet.xlsx
Waste Gravel EF.pdf
Bio-Oil Disposal EF Calc Sheet.xlsx
Bio-Oil Open Dump EF.pdf
Treatment of Food Waste.pdf
Treatment of Waste Wood.pdf
200015.xlsx
185-8201.pdf
185-8202.pdf
200016.xlsx
BATCH 200014, 200016, 200019 MRV(TAN + Density).pdf
185-8228.pdf
185-8231.pdf
200017.xlsx
BATCH 200017 MRV(TAN + Density).pdf
200018.xlsx
185-8273.pdf
200019.xlsx
BATCH 200014, 200016, 200019 MRV(TAN + Density).pdf
185-8286.pdf
200020.xlsx
BATCH 200020, 200021 MRV(TAN + Density).pdf
185-8345.pdf
185-8346.pdf
200021.xlsx
BATCH 200020, 200021 MRV(TAN + Density).pdf
185-8376.pdf
200022.xlsx
BATCH 200022 MRV(TAN + Density).pdf
185-8413.pdf
185-8414.pdf
200023.xlsx

Batch 200023 MRv(TAN+Density).pdf
June_July Diesel Conversions.xlsx
Security Oil 06072024.pdf
Security Oil 06102024.pdf
Security Oil 06182024.pdf
Security Oil 06242024.pdf
Security Oil 06282024.pdf
Security Oil 07022024.pdf
Security Oil 07062024.pdf
Security Oil 07102024.pdf
Security Oil 07152024.pdf
Univar 07022024.PDF
Vaulted July 2024 TOC.pdf
Vaulted June 2024 TOC.pdf
185-8201.pdf
185-8202.pdf
185-8228.pdf
185-8231.pdf
185-8273.pdf
185-8286.pdf
185-8345.pdf
185-8346.pdf
185-8376.pdf
185-8413.pdf
185-8414.pdf
KDHE Class V SW-12 Monitoring Form - July 2024.pdf
KDHE Class V SW-12 Monitoring Form - June 2024.pdf
ACS 2024 Second Quarter Report - SW12.pdf
Nov_Dec Diesel Conversions.xlsx
Secuity Oil 120524.pdf
Security Oil 102924.pdf
Security Oil 111824.pdf
Security Oil 112524.pdf
Security Oil 120324.pdf
Univar 101424.PDF
Univar 120524.PDF
Minifinery Electricity Consumption_ Nov-Dec Reporting Period_.xlsx
Copy of 10_08_24 Salt Delivery.pdf
Copy of 10_09_24 Salt Delivery.pdf
Copy of 10_24_24 Salt Delivery.pdf
200053 Vaulted Integrated Evidence.pdf
200054 Vaulted Integrated Evidence.pdf
200055 Vaulted Integrated Evidence (Truck 1).pdf

200055 Vaulted Integrated Evidence (Truck 2).pdf
200056 Vaulted Integrated Evidence.pdf
200057 Vaulted Integrated Evidence (Truck 1).pdf
200057 Vaulted Integrated Evidence (Truck 2).pdf
200058 Vaulted Integrated Evidence (Truck 1).pdf
200058 Vaulted Integrated Evidence (Truck 2).pdf
200059 Vaulted Integrated Evidence (Truck 1).pdf
200059 Vaulted Integrated Evidence (Truck 2).pdf
200060 Vaulted Integrated Evidence (Truck 1).pdf
200060 Vaulted Integrated Evidence (Truck 2).pdf
200061 Vaulted Integrated Evidence (Truck 1).pdf
200061 Vaulted Integrated Evidence (Truck 2).pdf
November 2024 AR-09 Well Report.pdf
October 2024 SW-12 Well Report.pdf
ACS 2024 Third Quarter Report - SW12.pdf
AR-09 Displaced Brine 11-20-2024.pdf
AR-09 Displacement Brine (12-4-24).pdf
Vaulted October 2024 TOC.pdf
Vaulted September 2024 TOC.pdf
IS2504Charm QA Checklist - JunDec24 Verification.xlsx
IS2504Charm_VerChecklist_RPJunDec24_2-14-2025.xlsx
ISO2504_Charm_RPJunDec24_VR_DRAFT_2-20-2025.docx
src_1J09YMYVT1S0MYNE_5146_19a62478-6089-4fd1-9551-d5b0e273fe65.pdf
src_1J09YMZ8Z1S08KH3_5145_868e56f8-6294-4c69-9e44-15a1af848192.pdf
src_1J09YMZ981S0W8CR_5142_bfe71fcc-7ccf-412d-aecc-1b2a9431c5b0.pdf
src_1J09YMZ9Y1S0FX8D_5148_8657192b-c2d7-4b63-83ad-992404eb9190.pdf
src_1J09YMZA41S03J42_5141_a79c2d8b-c750-4856-be94-75afeb817cdc.pdf
src_1J09YMZAK1S0Q6ZQ_5149_da0e1f02-0edc-4987-8ac8-b01c8f5f4821.png
src_1J09YMZXZ1S0AVVC_5155_4ed52cdc-48c9-4757-91be-f43a4a8085c3.pdf
src_1J09YN05Z1S0YGQ1_5154_cbf66a6d-e722-46fb-8a7e-31db5159eb67.png
src_1J09YN0AX1S05TEB_5158_937cc492-ef23-43d4-a735-aa28d10242e2.png
src_1J09YN0B01S0J5JP_5161_7e3dd154-4d6b-475f-986e-85b53f43fab0.png
src_1J09YN0DP1S0SFA0_5160_1f32ad3d-b5cc-4b1d-860e-f97c2b80e86d.png
src_1J09YN0ES1S0D45N_5157_4d902ee3-fec9-4ef5-8bf7-8795b75e1f5a.pdf
src_1J09YN0S01S00S1A_5163_e8627383-4087-4cd8-bfe6-19e54ac87b3e.png
src_1J09YN1121S0MDWZ_5162_3d654c22-daff-4a80-885e-dcd28314e572.png
src_1J09YN1ZR1S082RM_5168_2b450e91-c23e-4582-8731-71c8e469a3a9.png
src_1J09YN2211S0VQM9_5164_ebe90a71-d2cc-4d35-9b8c-b8a2123d9a33.png
src_1J09YN27V1S0FCFY_5165_f796a35d-761e-43a6-a503-4e43cf78667b.png
src_1J09YN2BT1S031BK_5169_948915e6-ed65-4315-915b-2c06dff796f8.png
src_1J09YN2DE1S0PP78_5167_3e7d7358-eba5-45a9-af5d-942eff91943c.png
src_1J09YN3001S0AB2X_5170_822548b1-f457-4099-93ae-b6245146ebc1.png
src_1J09YN3741S0XZYJ_5171_14b86bac-993e-4f05-8a54-3896fecb5248.png

src_1J09YN3SX1S0HMT7_5172_9f4a09b1-fcaf-45d3-9925-34d0d178b700.png
src_1J09YN4ZG1S059NW_5174_d6bf1641-081f-41f5-96d2-6efca22c58a1.png
src_1J09YN52X1S0RYHH_5173_fc48ab56-7f74-4728-be5f-3172f40278a5.png
src_1J09YN5DA1S0CKD6_5175_05662faa-573a-4a2b-9905-ec2c7a303aa6.png
src_1J09YN5ND1S0088V_5177_f16ed753-e8f9-49c4-a60e-3c2d3dd6f2fa.png
src_1J09YN5V11S0KX4G_5178_ab60df12-f00d-47ed-88de-e3a3ad0c6a57.png
src_1J09YN60D1S07J05_5179_dbb6c258-6574-4341-a436-3991e83555b2.png
src_1J09YN65E1S0V6VT_5183_3f464c46-3682-41a1-9019-0d6ed09a6c98.png
src_1J09YN7111S02GK4_5184_1e00b742-6ff4-448c-b9d6-6dae675ca62d.png
src_1J09YN7ME1S09TAE_5185_36a3ab0a-a741-4a63-9445-95ace6117826.png
src_1J09YN84B1S0H41R_5190_163275a4-c397-4dca-beb6-7df4a1795941.png
src_1J09YN8B81S04RXD_5191_0c372a84-316f-4503-90b3-049ada1fce98.png
src_1J09YNA4D1S0EAZ0_5209_408989ab-e667-4f7d-a972-a6534c8a7926.pdf
src_1J9M309841S0WF4K_5309_5bad944a-ddb7-46f8-928a-4b48fe632541.png
src_1J9M309WP1S0QDQJ_5310_7c491130-ddf6-4bb6-a41f-77780f42055c.png
src_1J9M30BCP1S0B2K7_5312_8a5839e9-1cd9-4594-ae70-db77fc1000a5.pdf
src_1J9M30BCQ1S0YQEW_5311_ea102848-d725-4166-af2f-bfc1a8bcd536.png
src_1J9M30BX91S0JCAH_5313_9ee3ecb8-50d8-4af1-898b-e927ccbf15bc.pdf
src_1J9M30CA01S06166_5314_b4a33aa0-194b-42ab-8603-b910429cb340.png
src_1J9M30FDK1S0AHTR_5387_1987575b-e3ff-4108-961e-d38419943578.png
src_1J9M30FGX1S0Y6PD_5388_84a18ddf-cfae-4dfa-9c3f-c5ae659bcba1.png
src_1J9M30GBD1S0HVJ2_5390_f2a36577-9b8a-407b-8f82-d740aa8423fe.png
src_1J9M30GF31S05GDQ_5389_764ed87b-08aa-4a98-9db5-fafa504b6e86.png
src_1J9M30GFM1S0S59C_5391_cca2d6da-7708-48c1-87a5-6e6485841f5e.png
src_1J9M30HBP1S0CT51_5392_b004155f-6def-40db-a17c-30f245b8ca70.png
src_1J9M30HZY1S00F0P_5394_d4fc00c3-8d9a-4b42-b2f1-85a6d786e27f.png
src_1J9M30HZY1S0M3WB_5393_86adb53a-2b42-4d6d-bdba-c006e0429ff7.png
src_1J9M30JCD1S07RR0_5395_082fa692-bf9a-4db2-9e40-4ad3b5d01af3.png
src_1J9M30JW61S0VDKN_5396_59ae5f2c-72b4-44f9-8130-0b6e41c672b7.png
src_1J9M30KB21S0F2FA_5397_83f9eabf-54e8-4d99-b4c4-ee59124b6efb.png
src_1J9M30KHD1S02QAZ_5399_41fd291a-ade6-4500-b780-78646c0126c9.png
src_1J9M30KMT1S0PC6M_5400_49692f14-9754-4fe5-a893-d5b841e8d725.png
src_1J9M30KTF1S0A129_5401_4a1f3162-56da-4fba-a109-8fe81817d683.png
src_1J9M30KXK1S0XNXY_5398_be6fa1e4-50e6-4295-ab87-62377744b49a.png
src_1J9M30M9K1S0HASK_5402_cda1ddda-adea-4f2a-b48a-c27bc0cf5118.png
src_1J9M30MBF1S04ZN8_5403_d0f7970c-2fbf-41e5-b12f-a8649c04a46d.png
src_1J9M30MPJ1S0RMGX_5404_1859678b-5b5c-47cc-835d-2a3dd46237fa.png
src_1J9M30NA81S0C9CJ_5405_2f18781d-c005-478f-a394-640041911321.png
src_1J9M30NWN1S0ZY87_5406_2c0431e0-0fd2-4e97-8af1-cefc89dc834f.png
src_1J9M30NZW1S0KK3W_5407_b42460c7-4f4c-42ca-9619-d794fce1c203.png
src_1J9M30PMJ1S077ZH_5410_4047dd28-2d0e-45c0-9eff-6b4eff9cc315.png
src_1J9M30PZS1S0EHPV_5409_a351e6fe-1439-4697-a8de-89bd455c52d8.png
src_1J9M30PZS1S0TWV6_5411_5245b0d3-d6f5-44f6-9cd6-741fd6299b5f.png

src_1J9M30Q661S026JG_5412_66f29b69-26ba-4ee0-af06-804dc6fd9c87.png
src_1J9M30RAM1S0NVE5_5413_79ebc68f-685b-43d5-ab6f-618aa9cf5581.png
src_1J9M30RXZ1S09G9T_5414_367eec36-8851-4eda-bc27-846271fcaa6f.png
src_1J9M30SZJ1S0X55F_5415_265badc9-7fe7-48ae-bc27-5eda01d110a2.png
src_1J9M30T091S0GT14_5416_7ffc15b8-0795-45e8-bec5-40f356d531ba.png
src_1J9M30TZD1S04EWS_5417_fd1e7f52-108c-42b5-803e-a34ca54448f4.png
src_1J9M30TZJ1S0R3RE_5418_b1e52248-0303-4fb6-afda-da3a3409efa3.png
src_1J9M30VDW1S0BRM3_5420_f279114b-5e4c-4256-9d1c-d839455038fc.png
src_1J9M30W0B1S0ZDFR_5419_12869338-c418-41da-b84e-76d4e8785012.png
src_1J9M30W3V1S0K2BD_5421_5368ace7-21f5-4e75-8cd8-d2ddb7dfdd6f.png
src_1J9M30WQ21S06Q72_5423_6aa7fda6-f416-4ba9-bde8-a86d2ce28187.png
src_1J9M30X3K1S0TC2Q_5422_5b50add2-6c08-40c4-a379-f06654f04fab.png
src_1J9M30XKP1S0E0YC_5424_b1607e96-80ac-441e-9a50-5958ddb32f76.png
src_1J9M30YRF1S01NT1_5425_3a6a60d5-28be-48b3-807c-90472cb86312.png
src_1J9M30YVE1S0NANP_5426_52160645-2127-4d79-ab9d-d171310fa48f.png
src_1J9M30Z931S08ZHB_5428_9933f176-f42b-428a-8b89-6dc588be52c1.png
src_1J9M310TJ1S03Y4A_5429_b7ced703-da92-4e4a-b687-ba7f2307cd52.png
src_1J9M310TJ1S0G98N_5430_617dff90-18b0-4f07-866b-6f54b1dd3cab.png
src_1J9M310TJ1S0WMD0_5427_e673a162-9107-4edb-afe0-b13164b7b246.png
src_1J9M311AA1S0QJZZ_5431_b8727940-94f2-45d5-90cb-2874340098e8.png
src_1J9M311NZ1S0B7VM_5432_7b4787ba-d761-4000-b870-498f8bb6e7e8.png
src_1J9M312631S0YWQ9_5433_65833fd8-1be0-4302-b01d-da22bc574ea7.png
src_1JH6PPKKK1S0RGE3_5765_da12b2bc-9fd4-49d1-8738-029a88165dbf.pdf
src_1JH6PPKKV1S0C59R_5767_52789782-8f1b-4a93-aa42-567b227062fd.pdf
src_1JH6PPM3G1S0KF12_5766_2a23e18f-7c42-4c02-8590-7e2903d7202d.txt
src_1JH6PPM3G1S0ZT5D_5764_e1100c3b-5fb8-4609-8a38-c2dbb36c858d.pdf
src_1JH6PPM5T1S073WQ_5763_38a5792e-90d4-48ad-b696-185429121da2.pdf
src_1JH6PPM631S0TRRC_5507_0b753d60-3a9d-4e24-ade8-47c0186c7c47.pdf
src_1JH6PPMYA1S0EDM1_5520_d46515e4-5d25-465a-bfe2-f8dd3d0be13e.PDF
src_1JH6PPNN41S0X12N_5510_dd984ac9-b656-4204-88dc-106b572b0337.pdf
src_1JH6PPRGZ1S0T7ZX_5538_e3c798ba-9f93-4a5e-a18c-f0543a1be6ea.pdf
src_1JH6PPSRS1S0N6JW_5688_d115af7f-327a-43f0-a101-b49b299ff0f0.xlsx
src_1JH6PPSSM1S08VEH_5689_5e55efc8-d3e7-4ce9-b91a-d51a533a87f8.pdf
src_1JH6PPSX01S03T1G_5690_0a1513cf-258b-44e4-a130-a87423f799ab.pdf
src_1JH6PPV0P1S0B3RT_5697_ecef72ba-5369-40bd-b2f3-28623a8ecb00.pdf
src_1JH6PPVPH1S0YRMF_5698_63f0a413-a434-48cd-9634-410b8405c817.pdf
src_1JH6PPW071S062BS_5705_f410369c-7a5d-4166-9ca2-ed16bcad3e4c.xlsx
src_1JH6PPW7Y1S0SQ7E_5716_c0367f34-beeb-4325-b14d-a30dcfc42793.xlsx
src_1JH6PPWPN1S010YR_5723_61202388-5d2d-4747-9243-0089d1068b31.pdf
src_1JH6PPWTY1S0MNTD_5724_5dfc7f72-ea94-45d3-8575-f2998b56dddb.pdf
src_1JH6PPXFY1S08AP2_5498_18816a76-baea-498f-befc-9d799876fd96.pdf
src_1JH6PPYAN1S0VZHQ_5499_82dfaae3-5a83-44d7-ac18-cfbf31996844.pdf
src_1JH6QA8TV1S0FD6F_5643_8fc52116-9f0a-4837-a5b9-1403940ac534.png

src_1JH6QA8YD1S0PPXS_5632_605ff594-aacb-45d8-87c2-b53a98d1a5a5.png
src_1JH6QA8ZD1S03224_5646_e07babd6-748e-4ae4-9ceb-b813470f1e34.png
src_1JH6QA98M1S0HNGR_5636_bb6a155e-5de9-4fb1-87ba-5ab54f2ba244.png
src_1JH6QA9961S0Y0N3_5637_9a3d0d97-da21-4ee1-a77d-0df9dd3ae089.png
src_1JH6QA9M01S05ACD_5648_ffcefba4-6034-486c-96f9-3a4be9cc1cf0.png
src_1JH6QAABD1S07JPP_5658_b1525c5b-dd98-4cd0-bdf8-d1afba05e3e9.png
src_1JH6QAACB1S0V7JB_5664_2dbbdcc1-6828-4788-8303-b44ccb94419b.png
src_1JH6QAAYP1S0EWE0_5665_34828f9a-0584-4e61-829b-c2dedb0088ce.png
src_1JH6QAC141S09V0Z_5669_8265ed0c-aa46-460e-9777-4aeda8809e80.png
src_1JH6QACYY1S0REFK_5681_afb9abe0-b5f5-4d37-aaf7-7d489b8e8d6b.png
src_1JH6QAD341S0C3B8_5685_845761ff-4fda-458d-b863-568aeff11348.docx
src_1JH6QAE771S0TPSW_5702_94d254ba-250f-4ea2-8ca6-da2559a98eb9.xlsx
src_1JH6QAEJ91S0EBNH_5710_efe3d683-c170-48f6-bd62-a58076d50ad4.xlsx
src_1JH6QAF3Q1S020H6_5718_6a68c2a2-1c97-45c7-a17a-1ea130c20a11.xlsx
src_1JH6QAFK71S0NNCV_5726_bf80fa9d-dc21-4ab5-b502-935244ebd933.pdf
src_1JH6QAG111S0GKZT_5769_dacc0bca-facf-4bfa-9253-43e94db3949f.pdf
src_1JH6QAG1R1S048VF_5768_dd6752fd-cd31-4548-b5c0-28b05d3175ac.pdf
src_1JH6QAG5G1S0QXQ4_5770_8f4917e6-0bd8-4701-9457-98af1c7e0d94.txt
src_1JH6QAGW81S0BJJS_5771_3664193c-39c9-4156-897f-4dbaf0b1e15a.pdf
src_1JH6QAKD91S0QCYN_5493_276ac975-693d-4f72-9758-be1e9c11466a.pdf
src_1JH6QAKN41S0B1TA_5492_ea3f6dc1-ea52-4a29-84d9-11d9b077d317.pdf
src_1JH6QAKX71S0YPNZ_5494_ccf23f85-6fe3-439c-8576-d1f175dca14e.pdf
src_1JH6QAM2S1S0JBHM_5501_8dc08955-571f-4440-8d6a-16d7603e6a7c.pdf
src_1JH6QAM431S060D9_5503_b4101c19-f843-4d71-b646-e8f0d97b5169.pdf
src_1JH6QAMMP1S0SN8Y_5508_49e3286d-a41c-44fe-9a13-ef9da2b9d3d1.pdf
src_1JH6QAN731S037AH_5511_98290c21-9870-4250-9e8a-6f750c068b35.pdf
src_1JH6QAN8T1S0MKVX_5523_2bec52d8-3d76-4ebb-bbc8-7f08347d703f.docx
src_1JH6QANJ31S088QJ_5519_96c90ae1-c298-437d-b82d-61b45f738b89.PDF
src_1JH6QANRC1S0VXK7_5528_68ecfeb6-75c0-4c64-98e8-84b09b2949f9.docx
src_1JH6QANY01S0FJEW_5524_757d5d69-f3cd-4771-a496-52dfdb19f297.docx
src_1JH6QAP2X1S0PW66_5525_a06298d1-92a5-4dce-b75c-6bbacf6e0d69.docx
src_1JH6QAPT31S0AH1V_5529_b1b4de8b-6a72-4902-b467-afcaa5b732a6.docx
src_1JH6QAQ771S0Y5XG_5534_bbc64602-42d8-4c5a-868a-a403c23888d1.docx
src_1JH6QAQMT1S05FMT_5549_023e40ee-263d-4aea-96a2-93931dbe1c7c.png
src_1JH6QAQRB1S0S4GF_5560_f4d21c3e-e3f3-4165-9780-40575b3ad973.png
src_1JH6QAQZE1S0CSC4_5562_f6cfa539-a267-4dcf-ad7b-49338967906f.png
src_1JH6QAR4B1S00E7S_5557_a1b53e89-79e0-474a-9eab-cc3acea4c8ea.png
src_1JH6QARDD1S0M33E_5565_bb3cd4e9-9ae4-42cc-82eb-7c4d4d621101.png
src_1JH6QAS0Y1S07QZ3_5571_eca69813-704b-4914-8b23-357c57cfd118.png
src_1JH6QAS581S0VCTR_5578_3b925b60-ae79-4d9e-aaf9-a72a85525a98.png
src_1JH6QASDA1S0F1PD_5569_8ada355e-26db-40e5-a0e3-c2f2dda72c12.png
src_1JH6QASNQ1S02PJ2_5572_8ab63765-ab49-4b6a-adb9-7764ae24f4ec.png
src_1JH6QASPK1S0PBDQ_5582_996af8b9-a5a1-4cbf-8ca2-e50203251f7b.png

src_1JH6QAT5C1S0A09C_5585_e0654a9f-1195-4ac6-a904-9c1782d9a3d9.png
src_1JH6QATBT1S0XN51_5583_6d47328d-7f6b-48da-a71a-c309036e8a44.png
src_1JH6QATQV1S0HA0P_5587_a4c33381-42ec-4888-982a-5bb200da8ad4.png
src_1JH6QAV111S04YWB_5602_7c94355a-a2ac-4f6f-a8a1-263545f0d4da.png
src_1JH6QAVC91S0RKR0_5608_687d5e0c-9b46-4c57-8507-46825a4e1142.png
src_1JH6QAVF01S0C8KN_5589_88e46afe-3e91-4aba-825d-8e9d973ac353.png
src_1JH6QAVVQ1S0KJAZ_5604_487c2fa1-282c-4f6a-8b6e-dc5b934d6a67.png
src_1JH6QAVW01S0ZXFA_5617_40de631d-c856-485b-9ef7-8a15e705b9f1.png
src_1JH6QG0H91S0767C_5634_209fd254-80e3-4029-b751-89ab0f36233f.png
src_1JH6QG0HN1S0KHBQ_5635_d4e67d0b-a569-4311-a475-1ae1e9eb192f.png
src_1JH6QG0K11S0TV31_5640_e5cb96fe-69d2-410e-a01a-f4fb146c7887.png
src_1JH6QG0QS1S0EFYP_5639_3797d622-4b57-4db0-9cb0-cea0fdfc023b.png
src_1JH6QG1061S024TB_5633_27aad466-2318-4aa4-83e3-a24b2e4ad86a.png
src_1JH6QG1711S0NSP0_5638_ecfb9750-879b-4364-9de9-2b515cf95d2d.png
src_1JH6QG1EZ1S09EHN_5642_e50c5d34-2226-4041-9918-593cc98b96a8.png
src_1JH6QG1JD1S0X3DA_5645_c468f1cd-ab45-45d3-bc4f-36a6d700db22.png
src_1JH6QG1RM1S0GR8Z_5644_112f80a2-d967-49df-b57f-8c57dfe16c20.png
src_1JH6QG1Y81S04D4M_5641_ace197a9-4a05-467a-a045-28a88986897b.png
src_1JH6QG1Z91S0R209_5650_b58a8191-08eb-49c4-919e-dd5637b95e22.png
src_1JH6QG2BA1S0BPVY_5647_fe0ea632-1026-4e1a-a8a8-c9c1183894c2.png
src_1JH6QG2K81S0ZBQK_5653_35bd5e35-102e-4ec2-a11f-a1a49bf40bd9.png
src_1JH6QG2QZ1S0K0K8_5651_ed23ed92-659e-42d6-8881-b9ef5808f956.png
src_1JH6QG2WP1S06NEX_5652_a3ae234d-ac25-42a2-bd26-991e25785e4d.png
src_1JH6QG3971S0TAAJ_5655_53c89a56-4f65-4985-97b6-faf5cba6e868.png
src_1JH6QG3GS1S0DZ67_5656_050a681f-2ca0-4320-a7ed-ad890c088907.png
src_1JH6QG3H41S01M1W_5654_682cd649-8071-443a-9aeb-f3073924d60b.png
src_1JH6QG4CR1S0N8XH_5657_f1807bb4-eec1-4a34-9277-2dda2e5c8c13.png
src_1JH6QG4DE1S08XS6_5659_b70787e1-2aa0-405a-b3ad-c72adc985cef.png
src_1JH6QG5861S03WC5_5660_f691b9f2-769f-4da1-9ee7-e178d82b55cd.png
src_1JH6QG5K61S0QH7T_5661_e21ab221-d739-4444-8956-fb9c9b110e13.png
src_1JH6QG5ZF1S0B63F_5663_31197961-7bd3-4b6c-a797-818795c873e5.png
src_1JH6QG6FH1S0YTZ4_5662_d5bf8695-7d9d-48a9-bacd-85fd0eccdd72.png
src_1JH6QG7031S064PE_5666_97c0adc0-963d-4215-a3b1-b1a9dbe5c45c.png
src_1JH6QG75X1S0SSJ3_5668_cce5dbdc-d71a-415d-ae5e-3ee07db07719.png
src_1JH6QG7E01S0DEDR_5667_a2b2d5f6-16e9-4044-8294-b81227aa8e22.png
src_1JH6QG8091S0139D_5670_2f76a995-9409-477f-9fd6-cc591385eb15.png
src_1JH6QG86V1S08D0Q_5672_6c8a03fb-715a-4bc8-b78e-8eae6292e78f.png
src_1JH6QG9Y01S0YA6N_5673_2c426c0b-2ce1-4c24-b46d-bd574dff554c.png
src_1JH6QGB0F1S0S8SM_5674_2a35aabc-7326-4f08-9143-c8918f89e2ca.png
src_1JH6QGBHH1S07W88_5683_b36cdfc1-5c8a-4dc7-9ca1-592b69ab8682.docx
src_1JH6QGCDP1S0F5ZJ_5679_fd957c6f-6c6c-4f29-90b6-9f925c9273eb.png
src_1JH6QGEEX1S001RF_5720_810191ff-ac45-4da4-ae0c-5564802aae9a.docx
src_1JH6QGEK21S07BFS_5727_4b15ec80-dbdf-41f6-8d4b-0c45795f1764.pdf

src_1JH6QGEZC1S0V0BE_5772_56e4ed71-05a5-4173-baf0-4b96772401bf.pdf
src_1JH6QGF7N1S0EN73_5773_4e06a468-c526-4b39-8f2a-f6b67d4a347e.pdf
src_1JH6QGFSF1S02A2R_5774_b07ce5e2-9586-4f57-a247-641a332d1807.pdf
src_1JH6QGFV11S0NYYD_5775_95c2e01b-2222-4d56-be1b-ca9dd9a84fa8.txt
src_1JH6QGGHM1S09KT2_5776_77fb2001-a854-431d-adf5-5dae8c157707.pdf
src_1JH6QGS6Y1S0303M_5675_b492f4fc-5bc1-4b07-addd-ecba24d7405c.png
src_1JH6QGT291S0A9TY_5676_e2eb705d-5caa-4397-9290-9bd59a51c93e.png
src_1JH6QGT3N1S0XYPK_5677_a696db42-6fb6-48c2-b46f-eec96974b8e3.png
src_1JH6QGT3V1S0HKJ8_5526_585fa1ca-b0cf-490e-a3e6-7107d2b5a340.docx
src_1JH6QGT7Q1S058DX_5527_13a3d389-39df-45f4-b4c5-0fa186c7093f.docx
src_1JH6QGT981S0RX9J_5678_d52c6d2d-1298-4a97-a006-339fd193abd3.png
src_1JH6QGTRY1S0CJ57_5530_f6a08cec-3b0b-4e44-b222-b94da1d38ef3.docx
src_1JH6QGTT41S0070W_5531_29117041-3af3-4752-bb19-397c097c4abd.docx
src_1JH6QGV721S0KVWH_5532_97a9406a-3a4a-4319-a415-8a62382cf0fe.docx
src_1JH6QGVE91S07GR6_5535_1c199e10-0e31-4fa3-aaf3-be96bf08d315.docx
src_1JH6QGVPK1S0V5KV_5537_e81aeb4e-00ff-44d6-b76f-1e3a4aec85ba.docx
src_1JH6QGVSM1S0ETFG_5533_030823fc-4a02-42da-ad21-122e5a299b30.docx
src_1JH6QGVTK1S02FB5_5536_f7157404-815b-4b09-9664-c36922847dd5.docx
src_1JH6QGWTK1S09S2F_5543_80627c50-edea-43ea-a84b-0ab57e7e5400.png
src_1JH6QGX3R1S0XDY4_5680_228c2bcf-5d5b-4ed3-a2b0-e20fe33dbf45.png
src_1JH6QGXB91S04QNE_5545_a3cb917a-0ebe-4551-9779-4938aba8518a.png
src_1JH6QGXCC1S0H2SS_5547_fdb8ee2b-d626-4e09-94b6-9c7101c174ca.png
src_1JH6QGXS21S0RCH3_5546_5d98e95e-765b-45d0-ada1-11ccd27dcd86.png
src_1JH6QGY2A1S0C1CR_5682_5e4f6092-3df0-43ba-a72d-a06cfa3106f8.png
src_1JH6QGYEQ1S0ZP8D_5550_00815d1c-817b-45cc-8f8b-695746e3eec3.png
src_1JH6QGYPY1S0KB42_5552_cd416fe2-a148-432b-9943-dca61d733f69.png
src_1JH6QGZ101S06ZZQ_5555_33c856ce-06d3-4b03-9055-4143bedeb2c3.png
src_1JH6QGZP41S0E9Q1_5558_c32eb699-f065-4b19-bf3e-75fbbdc88f03.png
src_1JH6QGZQZ1S01YJP_5567_1e7b3be2-8188-4102-a54a-72d2396a5310.png
src_1JH6QH01K1S0NKEB_5564_5fd871e5-97cb-4099-918d-618222ba8f01.png
src_1JH6QH04R1S098A0_5568_bf2d2425-3a91-4a83-bb98-238a4d5ae03f.png
src_1JH6QH0JH1S0WX5N_5573_b067c54b-c936-41cf-8b97-12cecca017ac.png
src_1JH6QH0MX1S0GJ1A_5570_d47f10e2-62f0-4dc4-8a18-d3701bbee746.png
src_1JH6QH12A1S046WZ_5575_f1f6302c-94d2-404c-8f2e-dde21a9e45e3.png
src_1JH6QH1561S0QVRM_5574_de6061f8-5ebc-40d4-b55b-5733dfa06619.png
src_1JH6QH16S1S0BGM9_5576_09080568-155a-489c-8a38-873c61de1a99.png
src_1JH6QH1MG1S0Z5FY_5577_e8b07ab7-d267-4352-ac08-4ad560896d3c.png
src_1JH6QH1W91S0JTBK_5581_f1761a1b-c267-4603-a428-075b00691350.png
src_1JH6QH23R1S06F78_5579_cc689bed-3d4f-4386-9369-f51939da67a8.png
src_1JH6QH29M1S0T42X_5580_2d40164a-c9fd-4966-b002-134d54c92229.png
src_1JH6QH2N71S0DRYJ_5588_8f7fc136-9a27-4a82-b623-169408bfae67.png
src_1JH6QH2NQ1S01DT7_5584_82ed12c2-04a1-4306-a3e8-caf17215d5c3.png
src_1JH6QH2RW1S0N2NW_5586_75193e4b-bc99-46a4-a3ef-cf809419ae05.png

src_1JH6QH2YT1S08QHH_5590_c976d4ed-6269-4501-a968-a941c2550c76.png
src_1JH6QH3CD1S0WCD6_5592_46a4afdf-4648-4716-ae20-06c49a9cc559.png
src_1JH6QH3NH1S0G18V_5591_0cdc914c-0156-4ed8-b96e-5d400c192921.png
src_1JH6QH41R1S03P4G_5593_d617f9a0-f0c8-4ea5-b0e7-4b685c9e3ae4.png
src_1JH6QH4BJ1S0QB05_5595_32578df5-ff03-495c-9dbc-148a6227d37d.png
src_1JH6QH4DS1S0AZVT_5594_c8c00726-3afc-4af0-84dc-17b9d11df299.png
src_1JH6QH4EK1S0YMQF_5597_baa029d2-36eb-49cd-b573-68b91ead3e16.png
src_1JH6QH4QH1S0J9K4_5596_57406364-9253-4c30-8c49-380fccf8a164.png
src_1JH6QH55G1S05YES_5600_74b75b90-9230-4f49-b56e-7e0659880d1c.png
src_1JH6QH5C21S0SKAE_5599_b958895a-e913-40bc-a485-578807f697f8.png
src_1JH6QH5D51S0D863_5598_9c45d7a2-7a99-4c5a-8c36-d4b41e4c2207.png
src_1JH6QH61M1S00X1R_5606_5309016b-62cf-4e25-867b-f58b745873ad.png
src_1JH6QH6891S0MHXD_5603_82aa0495-b35e-4828-8d97-1b07c828ad7f.png
src_1JH6QH6C11S086S2_5605_0642967d-8845-45bd-8024-f490eb199ed5.png
src_1JH6QH6E41S0VVMQ_5601_39ba0262-412a-46b5-b02f-8b205d73ea5c.png
src_1JH6QH6HW1S0FGGC_5607_d732490e-a1fe-4431-b039-c2473b60dca6.png
src_1JH6QH6RG1S035C1_5609_4e3528f6-c1b5-4f1d-927d-bde204db42dd.png
src_1JH6QH71T1S0PT7P_5610_4e7b6dc6-387d-4b02-9a46-dad0d03aab0b.png
src_1JH6QH7FB1S0AF3B_5612_cd0c6551-dfe8-43ed-ad47-11b3518e2885.png
src_1JH6QH7H61S0Y3Z0_5611_a671a55d-31ad-4961-9dae-eb4c8bc0f823.png
src_1JH6QH7KN1S0HRTN_5613_d1d0d3fc-cacc-4a1d-9f1e-08395a375edc.png
src_1JH6QH7RV1S05DPA_5615_f3c0003f-66d1-40c2-9dcc-92002dc77d4e.png
src_1JH6QH83E1S0S2HZ_5614_689339b1-ad55-4026-8fe8-0394b1510530.png
src_1JH6QH8591S0CQDM_5616_4dc9b568-e524-4ec1-bab2-09a9505c240d.png
src_1JH6QH89P1S00C99_5619_91622f5c-27eb-4471-8008-57c7a4e5efd8.png
src_1JH6QH8E41S0M14Y_5618_ce5ff80b-f8ca-41cb-bc90-501622ddf1a6.png
src_1JH6QH96W1S07P0K_5620_ba4706c9-fe88-4a87-b63f-19c96ecfef4d.png
src_1JH6QH9AR1S0VAW8_5621_63028fd0-9446-44e7-9512-4f0fc66c6cec.png
src_1JH6QH9NT1S0EZQX_5622_1cec9b87-9d53-4351-8ab6-d27ebe44ec54.png
src_1JH6QH9QM1S02MKJ_5623_fa75b89f-9f05-472a-9181-0a8b76692fdd.png
src_1JH6QHA741S0P9F7_5624_833a8b41-2436-46b3-be9e-26aa8212053e.png
src_1JH6QHAEV1S09YAW_5627_9991e55c-385b-452e-a72e-b23fa84c811b.png
src_1JH6QHAG71S0XK6H_5625_665a76d4-337c-4572-b335-28f0bb4b24bd.png
src_1JH6QHATS1S0H826_5626_78653d4e-9d6f-4f6b-97e9-bce1aad2e921.png
src_1JH6QHAXT1S04WXV_5628_35755b03-1218-4a72-8b03-d4d4826dceaf.png
src_1JH6QHBN91S0RHSG_5630_30b58aca-662b-483c-a157-2196d2302daf.png
src_1JH6QHBRZ1S0C6N5_5629_f016f74b-5403-40bb-b6b9-8f6afb82acc1.png
src_1JH6QHBWT1S0ZVGT_5631_24a1eefd-aded-4a80-889f-24c9ae8a19ba.png
src_1JH6QP22H1S0BENS_5728_fc8daa1b-45fc-4561-9ae3-b4da40c71981.pdf
src_1JH6QP29X1S0Z3HE_5729_45f83500-359c-4f39-b6a5-6c5d70fe5ba0.pdf
src_1JH6QP2HG1S0T24D_5779_812e9d61-8198-4d01-b28c-46b721213399.pdf
src_1JH6QP2TK1S01BVQ_5777_56b53382-d70a-4e96-bb43-9f9347f8ee1a.pdf
src_1JH6QP2VR1S0N0QC_5780_96251e6f-c055-4cf7-aee5-7f94feee13b4.txt

src_1JH6QP32M1S08NK1_5781_9fccf3cc-55c8-4ef3-8d4f-8bd7c68b585d.pdf
src_1JH6QP3D41S0FZAB_5778_dbaf83ee-5e49-4455-b8e1-86fb6a9b5cbd.pdf
src_1JH6R0EP91S0HACV_5730_5d448dbd-1b74-4b3c-8be0-1477f6cc06f2.pdf
src_1JH6R0FAT1S077JS_5785_4c806b21-7c16-4aee-bdbf-a75f8e91a706.txt
src_1JH6R0FHE1S0TWEE_5731_5952e38f-a0fd-4b22-b402-6e5dafb68df6.pdf
src_1JH6R0FJH1S0EHA3_5783_9aae7500-4d3d-428c-92d1-0ab8532fedd7.pdf
src_1JH6R0FRH1S0NV1D_5784_c8a98f53-6503-4dd1-a619-5f12f4061ff7.pdf
src_1JH6R0FRP1S0265R_5782_331c1941-d4b7-4588-ab1b-a9e3237a000b.pdf
src_1JH6R0G1W1S09FX2_5786_be6233b9-29bc-4cd2-9db4-ab2101d64260.pdf
src_1JH8BXY5D1S0C80J_5434_fcc64c82-8170-4b5f-89f9-3a9b0c301690.pdf
src_1JH8BXYQX1S0ZWW7_5437_0b658e89-5ee7-4728-9761-141831694c69.pdf
src_1JH8BXZ1S1S0KHQW_5435_28ef9b49-0863-4ac5-9bcd-d44f3590c7df.pdf
src_1JH8BXZ471S076KH_5436_3565eb26-e252-4208-a65f-2e4a223be9e6.pdf
src_1JH8BXZR31S0TVF6_5694_ad014081-7a19-4c20-819a-6be621da646d.pdf
src_1JH8BY06F1S0256G_5695_28c1a057-bf95-48dc-93c4-eec4766569c5.pdf
src_1JH8BY0FY1S0NT25_5438_5453a455-cc95-4136-8509-63bf464b430b.pdf
src_1JH8BY1EG1S0X3SF_5696_c4b4c250-82d6-431f-b387-4c9e35131e9a.pdf
src_1JH8BY1WW1S04DGS_5445_31e680de-9719-49c5-ae4c-4731bf2b65cf.pdf
src_1JH8BY2C61S0R2CE_5446_31ed0a54-6d90-4556-a4f1-b1a36730b1b6.pdf
src_1JH8BY2N11S0BQ83_5703_5a6b84a9-b24d-4f84-bb49-f9b8dc826ab2.xlsx
src_1JH8BY2TJ1S0ZC3R_5448_7bd824df-bf33-4eb4-8991-0be803f7264d.pdf
src_1JH8BY2Z21S0K0ZD_5447_db025d42-69b1-45dc-a996-57b3be73d8e0.pdf
src_1JH8BY3MH1S06NV2_5707_bced7f9c-43a4-4243-b309-3e2c0ea2424a.xlsx
src_1JH8BY4321S0TAPQ_5706_fbf3a4c1-73f6-426f-9134-1d0d5a0d6a98.xlsx
src_1JH8BY4641S0DZJC_5711_8c9858df-6a9d-40a6-84f0-7870138333d9.xlsx
src_1JH8BY4HZ1S01ME1_5712_4de3c0e0-3b3f-463c-91b1-f3e3b9a49ae7.xlsx
src_1JH8BY4R21S0N99P_5458_54762bd0-2d57-4dce-ab5f-2fb575fcd7e5.PDF
src_1JH8BY5171S08Y5B_5713_739a0da8-a7c0-4bfc-b07c-5f921c64ae93.xlsx
src_1JH8BY5PZ1S0WK10_5460_bcacf159-3976-4e1e-95a8-4dd73fdbbcb1.PDF
src_1JH8BY5Q61S0G7WN_5459_a76c25e3-d334-440b-8f69-13bf41a00251.PDF
src_1JH8BY6MV1S03WRA_5714_529370c6-1d4b-4090-95ac-b5a8626e6e3d.xlsx
src_1JH8BY73Q1S0B6FM_5457_f5babff2-430b-4f76-b1ec-a41abe98522e.PDF
src_1JH8BY7601S0YVB9_5717_373c272f-a183-4d5b-98e6-66084ab735f2.xlsx
src_1JH8BY7RT1S0JG6Y_5475_c4b828f6-b761-4dc3-b4f4-dedf67baa01b.pdf
src_1JH8BY82D1S0652K_5473_801939db-610b-484f-ae67-3263230f30d2.pdf
src_1JH8BY88E1S0SSY8_5474_8097625d-b6f8-482f-b04b-a3b840b82629.pdf
src_1JH8BY8EH1S0DESX_5476_c4e167e2-8407-4ec8-8a91-51a6a7724f82.pdf
src_1JH8BY9A71S013NJ_5478_2262cf17-fc27-4dbf-a4f3-83a93bfbcacb.pdf
src_1JH8BY9PG1S0MRH7_5482_2bac8769-4dc8-4b19-accf-07f7e94f2f24.pdf
src_1JH8BY9V41S08DCW_5483_8f562cb5-0043-479d-b648-297be35cc7e1.pdf
src_1JH8BY9ZN1S0W28H_5754_c0126e65-2c20-44dd-8b90-4b7e3ce0ca18.pdf
src_1JH8BYA481S0FQ46_5755_bfacce98-b137-441a-b4a9-364e6a583e20.pdf
src_1JH8BYAD11S03BZV_5757_72e4f507-1f15-4591-a245-252f44c3835c.txt

src_1JH8BYAFR1S0Q0VG_5756_20fdd770-cc21-46b1-99b4-c31110450a29.pdf
src_1JH8BYBA71S0ANQ5_5758_33877839-c65e-43a5-87c1-10656df4ef28.pdf
src_1JH8C05HR1S0K8JF_removal_745_graph.pdf
src_1JH8C4HX11S08WK0_5505_822920e6-eba3-4dc2-b000-fa3b5b31fd74.pdf
src_1JH8C4J8Q1S0WHEN_5760_4c146a16-62ca-4de1-8dab-11d03cbbcb41.pdf
src_1JH8C4JCE1S03V5Z_5762_13f4681b-22f4-49af-b1ea-5ed963934b3e.pdf
src_1JH8C4JDX1S0QG1M_5509_a1bfb106-8bce-487d-99c6-15daab568a99.pdf
src_1JH8C4JEE1S0B4X9_5761_bc0abf05-25bc-422d-840a-383af9fce880.txt
src_1JH8C4JEQ1S0YSRY_5504_d0495d7c-62b6-427b-b6d2-b41774f1912c.pdf
src_1JH8C4K921S063G8_5506_2bb60805-7769-434d-bf54-2098df2a9c0e.pdf
src_1JH8C4KPV1S0DD7J_5512_081bcc13-219a-4725-9b6c-9715a7ec4485.pdf
src_1JH8C4Q2A1S0M66D_5691_278c19b5-8b30-4baf-bffe-742df889da36.pdf
src_1JH8C4QH91S07V22_5692_8e1a9ae5-9556-46a8-bad2-c73965e55925.pdf
src_1JH8C4QXP1S0F4SC_5693_94385fee-3d0d-42c2-b70c-018e6f4e42a0.pdf
src_1JH8C4RYC1S0HD3N_5704_cb3c94e7-d035-4503-b77f-6032849dea45.xlsx
src_1JH8C4S3F1S051ZA_5708_80e070f4-68da-4cb8-9170-7c3365ce6ab6.xlsx
src_1JH8C4SNK1S0RPTZ_5715_b6a466e5-c3e0-4d4b-be89-abfc0945cae8.xlsx
src_1JH8C4SRA1S0CBPM_5709_621a9fe0-b127-4555-813f-f1c4e81d237c.xlsx
src_1JH8C4TDQ1S0KNDY_5742_f37270c4-4e6d-4468-81e6-1750acde7cf1.xlsx
src_1JH8C4V1J1S0EM0X_5744_93d9a91b-83b5-44ae-b90d-874906edec32.xlsx
src_1JH8C4VGQ1S028WJ_5745_ff1ad4c2-260b-4646-9019-9ba41f7b875f.xlsx
src_1JH8C4VQJ1S0NXR7_5747_8fae3ff2-917f-4aea-8f4d-ec496eee1081.pdf
src_1JH8C4W7Z1S09JKW_5500_6a54525d-784a-49eb-bf6d-72dc39cdea6b.pdf
src_1JH8C4WFH1S0GWB6_5502_71f8d7f9-6740-42ce-9573-21e97b702c8f.pdf
src_1JH8C4WK21S04H6V_5759_a6ca1a07-6160-4fab-a07b-6c1ad58bd0e7.pdf
src_1JH8C5R8Q1S00A3V_removal_746_graph.pdf
src_1JH8C9D881S0AMGZ_removal_747_graph.pdf
src_1JH8CEZRY1S0HSC1_removal_748_graph.pdf
src_1JH8CP1F41S0SCDZ_removal_749_graph.pdf
src_1JH8CTW6B1S0E0N4_removal_750_graph.pdf
src_1JH8CXV241S0JMKK_removal_751_graph.pdf
src_1JH8CZBRA1S0ZT1Z_5733_3a0e0eb6-e60b-49f4-a3c9-254cdf62284c.pdf
src_1JH8CZBT91S0TRMY_5732_0012fc2b-e6fd-421a-be8c-b08652f2835f.pdf
src_1JH8CZCFZ1S04APH_5789_9e861f13-bd55-4f93-a7b9-8ea7c367ca69.pdf
src_1JH8CZCN71S0QZJ6_5788_ef431fc7-3a6f-4b8a-819f-1590324bd417.pdf
src_1JH8CZCT61S0Z99G_5787_79f31bde-755e-4cc8-ba3c-8a977373750f.pdf
src_1JH8CZD8F1S06K0T_5791_6a1e58d1-0b68-49aa-a02e-cbf407da5851.pdf
src_1JH8CZDKN1S0DWR4_5790_2f802749-01a7-4b5e-8179-cbe775a92bb6.txt
src_1JH8CZY6N1S04TFR_removal_752_graph.pdf
src_1JH8D13FZ1S0TB6C_5794_24f11e13-8aa5-4dd2-87ee-a045489e9ba3.pdf
src_1JH8D13GJ1S0E021_5796_5d7d3d76-0272-4f89-807a-ecadd84471e6.pdf
src_1JH8D13J71S01MXP_5793_1ad2e456-5bc7-45eb-955f-53b16ae6a9d0.pdf
src_1JH8D13WQ1S08YN0_5792_8aa786f9-bdf9-41e9-8779-b2d4e80b4c20.pdf

src_1JH8D143B1S0WKGN_5795_848e8190-8cf4-4cfe-a37c-1d10e67f953c.txt
src_1JH8D148E1S0QJ3M_5748_42fcb6d7-355f-456c-baca-6ec78401cdf2.pdf
src_1JH8D14NP1S065J8_5749_86642b41-f5eb-449d-a875-2658587a86f5.pdf
src_1JH8D1NVK1S024PX_removal_753_graph.pdf
src_1JHJYYMAD1S0NZ06_5797_c18ed3e9-812b-4d8c-ab15-92c2023b1cc7.pdf
src_1JK6DPNSG1S0M746_5798_a4809608-f4e3-4cb1-b1ae-28060fd7ca73.pdf
src_1JKH02KAE1S0HAD5_5801_d61faae7-8b07-4151-857b-66222c9e9174.pdf

APPENDIX 2: VERIFICATION AUDIT FINDINGS AND RESPONSES

ID No	Type	Finding / Issue	Audit Evidence	Protocol Reference	Required Action	Conclusion / Resolution
1	Forward Action Request	Charm injection site (Vaulted Deep operated) is installing new gas monitoring equipment to better identify gas production at the wellhead / brine holding tanks. Updated information should be provided	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	3.1.3, 3.2 (Biomass or Bio-oil Storage in Salt Caverns Module)	Provide specifications, design, installation information, calibrations, and operational data for the new gas monitoring equipment when it becomes available and for relevant reporting periods	According to Charm, Vaulted has not yet installed this equipment. The requested information will be provided once installation is complete
2	Forward Action Request	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing. Has this been completed yet?	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1		Please provide AECN testing report. Note, likely not applicable for this reporting period, but shouldbe updated when possible	Testing not completed yet. Remains as FAR
3	Clarification	Biomass feedstock sourcing was confirmed at validation. Because of its critical nature for compliance, we feel it should be confirmed for each verification reporting period - demonstrating no change in feedstock sources since validation		7.2	Please provide confirmation of biomass feedstock sourcing and that it is not different from validated sources (mill wastes)	Included as FAR for future verifications
4	Forward Action Request	Review and update the KS grid electricity EF annually based on changes to grid mix and any eGrid updates	Minifinery Electricity Consumption_Nov-Dec Reporting Period	7.4.3	Please review and update as needed going forward	Maintain as FAR – confirmed by Charm for review at next eGrid revision.
5	Omission	Documentation for BCUs EC3, EC4, and EC5 not provided		Transportation Emissions Module 4.1	Provide relevant data and approaches taken to confirm eligibility	Complete. Files supplied.
6	Clarification	Unclear if the PDD has been updated to reflect changes during this reporting period	GHG Statement	6.1	Clarify if an updated PDD exists	Complete. PDD has not been updated but all changes are noted in GHG statement.

7	Nonconformity	Some removals only have 1 associated C content sample, not 3 as required by module	[External] 2024-12-06 Response to Charm on Proposed Revisions to Injectate Testing Program.docx	Bio-oil Geological Storage module 7.4.1.1.1	Provide C content data for production batches with fewer than 3 samples or justification that there is minimal variability.	Complete. Injection batches only need to be sampled once. Production batches must be sampled 3 times, if sampled. Confirmed by Isometric Dec 6, 2024

APPENDIX 3: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	**Criteria Met?**	**Evidence / Notes** *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Lily Schacht
Carbon Removal Verification Engineer, 350Solutions

EDUCATION:
MS, Environmental Engineering, University of Wisconsin – Madison, 2019
BS, Chemical, Energy, and Environmental Engineering, Washington University in St. Louis, 2017

EXPERIENCE SUMMARY:
Lily Schacht is an Environmental and Chemical Engineer with experience in process engineering, environmental chemistry, analytical methods, and life cycle analysis (LCA). At 350Solutions, Lily works on verifying carbon dioxide removal (CDR) technologies, with a focus on mineralization-based pathways, including enhanced weathering, direct air capture, and ocean alkalinity enhancement. Previously, Lily led agronomic research at an enhanced weathering CDR supplier where she organized field trials across multiple states to quantify carbon removal and crop yield changes after rock application. Before that, Lily developed rapid prototyping instrumentation to optimize a biomineralization process in concrete production and aided in scaling up the process to pilot-scale. Lily also built environmental impact models for process variable sensitivity analysis of demo-scale manufacturing processes. These models were used to drive the direction of research and development to minimize product life cycle impacts. Throughout her career, Lily has developed analytical chemistry methodologies for both liquid- and solid-phase analyses.

RESEARCH AND PROFESSIONAL EXPERIENCE:
March 2024 – Present: Carbon Removal Verification Engineer, 350Solutions
Verify CDR technologies on behalf of registries and the XPRIZE Carbon Removal challenge. Specializes in mineralization pathways, including mineralization kinetics, measurement methods, and open-system modeling.

Nov 2022 – Feb 2024: Researcher, Lithos Carbon
Quantified carbon removal rates and agronomic impacts of enhanced weathering on cropland across six US states; Evaluated chemical analysis methods for precision relative to cost.

Nov 2023 – Jan 2024: Independent Consultant, Keel Labs
Built an environmental impact model to evaluate potential material and process changes; Recommended areas for reducing material usage up to 80%

Dec 2020 – Oct 2022: Research Scientist II
Guided experimentation and data analysis throughout all R&D teams to inform techno-economic analysis (TEA) and LCA; built and led the carbonate biomineralization prototyping workstream for rapid iteration; developed real-time measurement techniques for critical process parameters in solid state

SELECTED PUBLICATIONS & PRESENTATIONS:

- **Schacht, L.**, Baum, M., Liu, H., & Yap, M. (2023) Scaling Enhanced Rock Weathering: Agronomic Impacts at Field-Scale [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO.
- **Schacht, L.** and Ginder-Vogel, M. Arsenite Depletion by Manganese Oxides: A Case Study on the Limitations of Observed First Order Rate Constants. *Soil Syst.* 2018, 2(3), 39. https://doi.org/10.3390/soilsystems2030039

Guy Ingram-Hardwick
Carbon Removal Verification Engineer, 350Solutions

EDUCATION:
MEng Materials Science and Engineering, Loughborough University, UK - 2022

EXPERIENCE SUMMARY:
Guy Ingram-Hardwick is a Materials Engineer with experience in process engineers, MRV protocol development, experimental design and life cycle analysis (LCA). At 350 Solutions, Guys effort center on validation and verification of varied carbon removal pathways. Guy began his experience in the carbon markets directly after graduating in 2022 working to develop an LCA model for Brilliant Planet, a marine carbon removal company growing and burying microalgae for carbon sequestration. Once completing the LCA model, Guy managed the third-party verification and co-authored a paper awaiting publication documenting the findings. Guy also co-authored the MRV methodology for Brilliant Planet before managing its adoption with carbon market registries and developing the relevant PDD documentation. Guy Master's thesis in degradation mechanisms for biodegradable polymers provided him experience in experimental design and execution which lead to his experience at Brilliant Planet in conducting field trials aimed at displaying the long-term permanence of the carbon removal system. Prior to his experience at Brilliant Planet, Guy worked as a process engineer at Pirelli's rubber compound manufacturing plant in Burton-on-Trent with a focus on continuous improvement.

RESEARCH AND PROFESSIONAL EXPERIENCE:
January 2025 – Present: Carbon Removal Verification Engineer, 350Solutions
Verify CDR technologies on behalf of registries and private investment ensuring high quality and meaningful climate impact.

Jul 2022 – Dec 2025: LCA and MRV Associate, Brilliant Planet
Quantified the carbon removal efficiency of the Brilliant Planet system across a variety of engineering designs using LCA. Developed the proprietary MRV methodology and PDD as well as setting up a novel experimental design to display the permanence of the stored carbon.

July 2019 – September 2020: Process engineer, Pirelli
Completed data analytics to drive continuous improvement for increasing efficiency, safety and rubber compound quality.

350Solutions, Inc. Corporate Experience

350Solutions serves as an independent expert in cleantech, low carbon, and environmental technologies. We provide an unbiased assessment of innovative technologies. 350Solutions is accredited through ANAB under ISO 17020 as an independent inspection body to provide independent technology evaluation services using the ISO 14034 ETV process. 350Solutions is an approver verifier for both the Isometric and Puro. Eart carbon removal registries. In addition, 350Solutions staff include a Certified Measurement and Verification Professional (CMVP for IPMVP) and a North Carolina Registered Professional Engineer (P.E.).